MAT102 - Week 3: Mathematical Induction

Slide 1: The Principle of Mathematical Induction (PMI)

To prove that a statement $P(n)$ is true for all $n \in \mathbb{N}$:

1. Base Case: Prove $P(1)$ is true.
2. Inductive Step: Assume $P(k)$ is true for some arbitrary $k \in \mathbb{N}$. Prove that $P(k+1)$ is true.

Slide 2: Example Proof

Prove that $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.